Aritro and the Group Chat

The group chat was once buzzing with messages, reactions, memes, and jokes every nanosecond, like the peak of any group chat. Alas, the glory days are now over; only Aritro sends messages in the chat now. The others have a life now!

There are N inactive friends in the chat. Aritro sends messages one by one, and each message is assigned an **ID** equal to the number of messages sent so far. So his first message has **ID** 1, his second has **ID** 2, and so on.

A friend can come online at any point. When friend f comes online, he reads all messages that have been sent so far, and then immediately goes offline. If Aritro sends more messages after that, friend f will not see them unless he comes online again.

You will be given Q events in order. Each event is one of the following:

1. Aritro sends a message. It is assigned the next available ID.
2. Friend f comes online and reads all messages sent so far. He then goes offline.
3. How many friends have read the message with ID s?

Note: A friend reads the message with ID s if and only if he came online at a time when **at least** s **messages had already been sent**. For every type 3 query, it is guaranteed that at least s messages were sent beforehand.

Input Format

Input is given in the following format:

- line 1: N Q
- line $1 + i$ ($1 \leq i \leq Q$): the i-th query, where the first value is its type t, which is one of the following:
  - `1`
  - `2 f` ($1 \leq f \leq N$)
  - `3 s`

Output Format

- For every query of type 3, print the number of people who have seen the message with ID s

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $1 \leq Q \leq 5 \times 10^5$

Subtasks

Subtask	Score	Additional constraints
1	10	$N = 1$
2	20	$N, Q \leq 5000$
3	30	All type 3 queries appear after all other queries
4	40	No additional constraints

Examples

Example 1

Input:

```
4 11
1
1
2 1
3 1
1
1
2 2
3 1
2 3
2 4
3 4
```

Output:

```
1
2
3
```

Note

Query types:

- If $t = 1$, Aritro sends a message in the group chat.
- If $t = 2$, another integer f is given ($1 \leq f \leq N$). Friend f comes online and reads every message sent beforehand.
- If $t = 3$, another integer s is given. Aritro asks how many friends have seen the message with ID s.

Explanation of Example:

- **Query 1:** Aritro sends a message. It is assigned id 1.
- **Query 2:** Aritro sends a message. It is assigned id 2.
- **Query 3:** Friend 1 gets online. He reads the messages with id 1, 2.
- **Query 4:** Aritro asks how many friends have seen message with id 1. Only friend 1 has seen it. So, the answer is 1.
- **Query 5:** Aritro sends a message. It is assigned id 3.
- **Query 6:** Aritro sends a message. It is assigned id 4.
- **Query 7:** Friend 2 gets online and reads the messages 1, 2, 3, 4.
- **Query 8:** Aritro asks how many friends have seen message with id 1. Friends 1, 2 have seen it. So, the answer is 2.
- **Query 9:** Friend 3 gets online. He reads the messages with id 1, 2, 3, 4.
- **Query 10:** Friend 4 gets online. He reads the messages with id 1, 2, 3, 4.
- **Query 11:** Aritro asks how many friends have seen message with id 4. Friends 2, 3, 4 have seen it. So, the answer is 3.